

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Abrasive pad, Part Number 3823258, or equivalent
- Sealant, Part Number 3164067, or equivalent
- Sealant, Part Number 3164070, or equivalent
- Guide, Centering, Part Number 5395409

Additional Service Items

- Feeler gauge or shim stock
- Fine crocus cloth
- Loctite™ 262 or 271
- Plastic hammer

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- If equipped with a wet flywheel housing, drain the oil from the flywheel housing by removing the plug in the bottom of the flywheel housing.
- Remove the starting motor. Reference Procedure 013-020 in Section 13.
- Remove the transmission and all related components, if equipped. See equipment manufacturer service information.

- Remove the flywheel, if equipped. Reference Procedure 016-005 in Section 16.
- Remove the flexplate, if equipped. Reference Procedure 016-04 in Section 16.
- Remove any components attached by the original equipment manufacturer (OEM) to the flywheel housing, such as mufflers, shift mechanisms, air filters, and so on. See equipment manufacturer service information.
- Remove the flywheel housing. Reference Procedure 016-006 in Section 16.

Remove

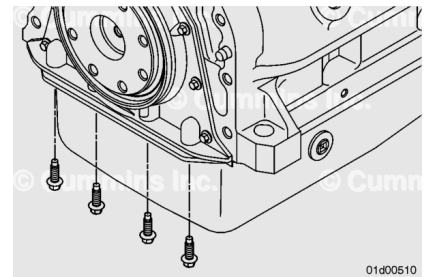
⚠ CAUTION ⚠

Use extreme care when releasing the lubricating oil pan gasket from the rear cover to prevent damage to the gasket.

Remove the four lubricating oil pan mounting capscrews that secure the oil pan to the rear cover.

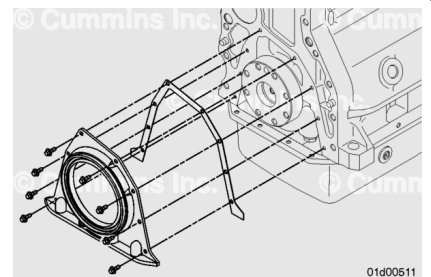
Insert the feeler gauge or shim stock between the rear cover and the oil pan gasket.

Move the feeler gauge back and forth to release the gasket from the rear cover.

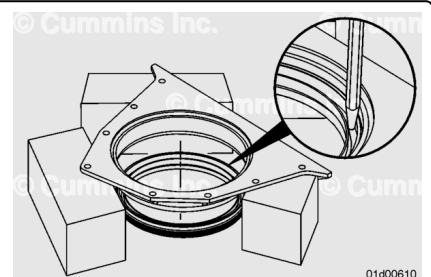


Remove the capscrews from the rear cover.

Remove the cover from the crankshaft flange



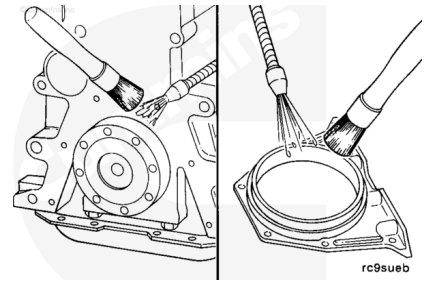
Support the rear seal carrier on a flat work surface with wooden blocks. Use a suitable punch and hammer to drive the old seal out of the rear seal carrier.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



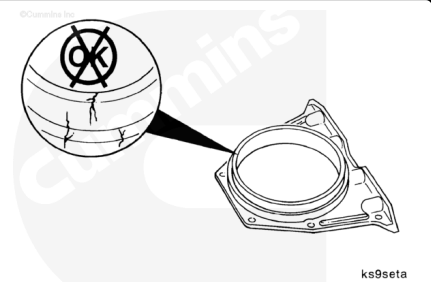
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the gasket surface of the cylinder block and rear seal carrier with solvent.

Dry these areas with compressed air.

Inspect the rear seal carrier for cracks or other damage. Replace the rear seal carrier if any damage is found.



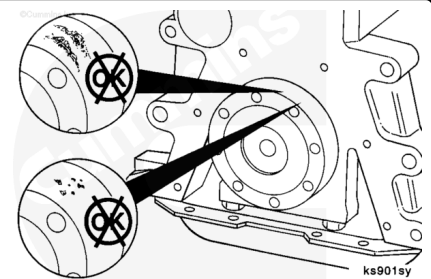
Use a fine crocus cloth to remove any rust or other deposits from the crankshaft flange.

Use a clean cloth and solvent to clean the crankshaft flange.

Inspect the crankshaft flange for nicks or burrs. Use an abrasive pad, Part Number 3823258 or equivalent, to remove any minor damage.

Note : If the cranksahft has excessive wear, a service wear sleeve is available.

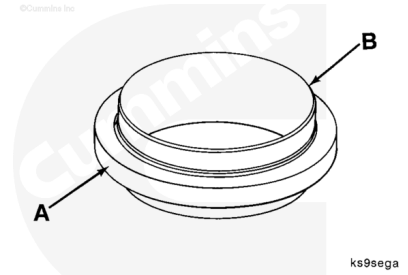
For engines equipped with a lip style rear crankshaft seal, inspect the crankshaft seal contact area for a wear groove. If the groove is deeper than 0.25 mm [0.010 in], a wear sleeve and oversize seal **must** be used. Reference Procedure 001-067 in Section 1.



Install

Note : For engines equipped with a wet flywheel housing, make sure to replace the rear crankshaft seal with the correct rear seal. Rear crankshaft seals for wet and dry flywheel housings may **not** be the same.

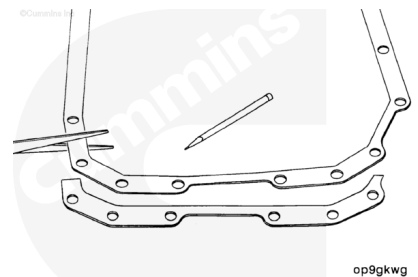
The new rear crankshaft seal (A) will come installed on a seal pilot (B). The seal **must** be left on the seal pilot while installing the seal onto the nose of the crankshaft. This will keep the lips of the seal from being damaged during installation.



Service Tip:

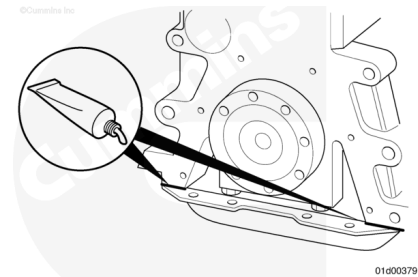
If the oil pan gasket was damaged during removal of the real seal carrier, remove the damaged gasket and:

- If previously equipped with a paper oil pan gasket, use the old gasket as a pattern and cut a section of new gasket to the same size. Use a light coat of sealant, Part Number 3164067, or equivalent, to hold the gasket in place.
 - If previously equipped with a sealant **only** oil pan gasket, apply a bead of sealant, Part Number 3164070, or equivalent, to the oil pan flange.
 - If previously equipped with a suspended oil pan, the oil pan **must** be removed and a new gasket installed.
- Reference Procedure 007-025 in Section 7.



Note : It can be necessary to trim the rear seal carrier gasket so it is even with the oil pan mounting surface. Test fit the gasket prior to installing.

Apply a thin bead of sealant, Part Number 3164067, or equivalent, at the intersecting joint of the oil pan and cylinder block prior to installing the rear crankshaft seal carrier. The rear crankshaft seal carrier **must** be installed within 10 minutes of applying the sealant.

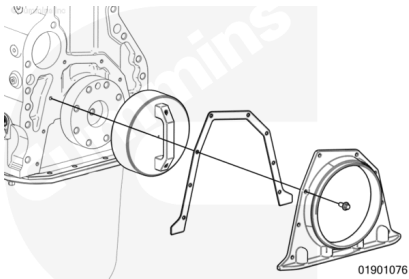


Install the crankshaft seal carrier using the centering guide, Part Number 5395409.

Use the centering guide tool to align the crankshaft seal carrier by using the end of the crankshaft as the pilot.

Install the rear crankshaft seal carrier and gasket. Apply Loctite® 262 or 271 to the mounting capscrews.

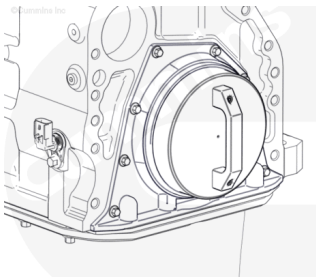
Loosely tighten the rear seal carrier to the block.



⚠ CAUTION ⚠

Do not push or force the cover in any direction. This can cause an irregular seal lip position after seal installation. An engine oil leak will result.

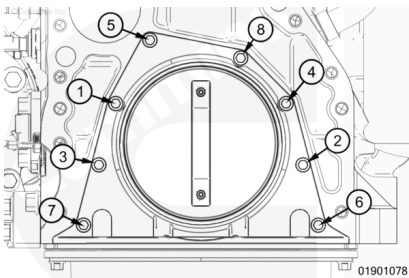
Align the rear cover even with both sides of the oil pan rail on the cylinder block.



Tighten the rear cover capscrews.

The chart below shows the proper torque value when using a 9.8 or a 10.9 grade bolt. The grade is embossed on the top of each bolt.

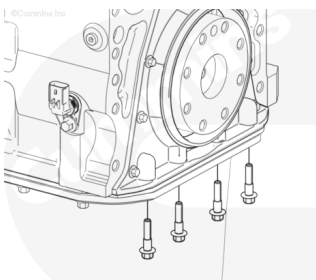
Apply Loctite® 262 or 271 to both part numbers below during installation.



Rear Seal Carrier				
Bolt Number	Bolt Size	Bolt Class	Torque	Loctite®
3913638	M-6	9.8	10 N•m [89 in-lb]	Yes
3991306	M-6	10.9	13 N•m [115 in-lb]	Yes

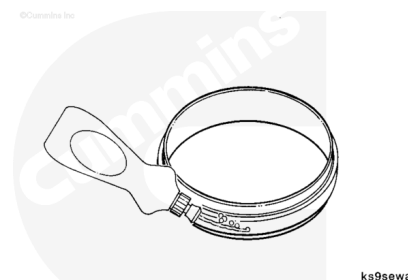
Install the four rear lubricating oil pan mounting capscrews into the rear cover.

Torque Value: 28 n•m [248 in-lb]



Remove the centering tool.

To aid in installation, the lubricating oil lip seal requires the application of a mild soap or P-80, Part Number 3824878, on the outside diameter of the seal.

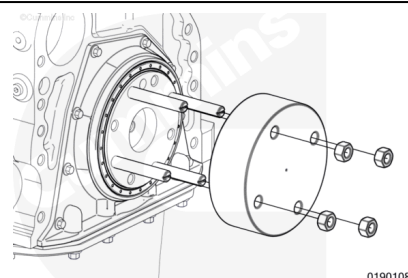


Place the new rear crankshaft seal, with tapered end of the crankshaft seal pilot over the crankshaft nose, and slide it by hand toward the crankshaft seal carrier.

Remove the crankshaft seal pilot.

Use Rear Crankshaft Seal Installation Kit, Part Number 5395259, to install the rear crankshaft seal. Refer to Procedure 001-024 in Section 1.

(/qs3/pubsys2/xml/en/procedures/100/100-001-024.html)



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the flywheel, if equipped. Reference Procedure 016-005 in Section 16.
- Install the flexplate, if equipped. Reference Procedure 016-04 in Section 16.
- Install the starting motor. Reference Procedure 013-020 in Section 13.
- Install the transmission and all related components, if equipped. See equipment manufacturer service information.
- If equipped with a wet flywheel housing, fill the flywheel housing with oil. See equipment manufacturer service information.
- If previously removed, install any components attached by the OEM to the flywheel housing, such as mufflers, shift mechanisms, air filters, and so on. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for proper operation.

